coursera ← Back **Computers and Logic**
Graded Assignment • 15 min

Search in course **Search**

English · Due Nov 4, 11:59 PM IST · English

≡ **Hide menu**

Understanding the Difficulty of Problems

- **Video:** Hard Problems for Computers
  22 min
- **Video:** Machines and Logic
  25 min
- **Discussion Prompt:** Propositional Logic
  10 min
- **Graded Assignment:** Computers and Logic
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Next item →

Computers and Logic

1. Among Polya's problem-solving heuristics is the suggestion "Look for a related problem that you know." Why might this be an interesting or challenging suggestion for a computational problem solving system. **1 / 1 point**

( ) Most problems are one-of-a-kind, and aren't, in fact, related to any other problems.
(•) Pursuing this heuristic would involve tackling the notion of similarities or analogies between various (superficially distinct) problems. This would be an interesting research project!
( ) People do not in fact use Polya's heuristic for problem-solving.
( ) There is rarely enough computer memory available to create a "repository" of problems.

✓ **Correct**

Review Learning Objectives

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Assignment details

Due Nov 4, 11:59 PM IST **Attempts** Unlimited **Try again**

Submitted Nov 1, 11:41 PM IST

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2. The "rotating-quarters" problem is difficult to approach via the same methods as (say) Rubik's Cube. Which of the following reasons is most relevant to this difficulty? **1 / 1 point**

(•) Finding a solution appears to involve elements of mental imagery or mental simulation.
( ) The behavior of coins on a table involves a great deal of "common-sense" knowledge.
( ) The problem is not well-defined enough to have a clear solution.
( ) There are many ways to solve the problem; it's not clear which one to use.

✓ **Correct**

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3. The "Teddy Roosevelt" problem is difficult to approach via the same methods as Rubik's Cube. Which of the following reasons is most relevant to this difficulty? **1 / 1 point**

( ) The problem is given in words rather than mathematical equations.
(•) There is a potentially (extremely) wide range of real-world or common-sense knowledge involved in answering the question; so the problem is not self-contained as many puzzles are.
( ) The problem is not well-defined enough to have a clear solution.
( ) We can only answer the problem with a statement of probability; we can't be certain of the answer.

✓ **Correct**

4. Consider the "10 coins in three cups" problem given at: https\://curiosity.com/topics/can-you-solve-the-3-cups-10-coins-logic-puzzle-curiosity **1 / 1 point**

What might make this difficult to solve with a computational system?

(•) A "standard" search program assumes a particular representation of the problem, while this particular problem involves finding a creative reconsideration of the problem statement itself.
( ) There is a tremendous amount of common-sense knowledge about (e.g.) coins and cups required to solve this problem.
( ) Since the problem can't be solved, a computer program is bound to fail at it.
( ) This problem is not well-defined enough to have a clear solution.

✓ **Correct**

5. Consider the "monkey climbing a rope" problem given at: **1 / 1 point**

https\://activityworkshop.net/puzzlesgames/monkey/index.html

What makes this problem difficult?

( ) The problem involves an unavoidable degree of probability or uncertainty.
(•) The problem involves elements of physics knowledge and (most likely) mental simulation and imagery.
( ) The problem has an astronomically large problem-space representation.
( ) There is no solution to the problem.

✓ **Correct**

6. Consider the "sand timers" problem (Problem 4) at the following website: https\://www.analyticsvidhya.com/blog/2016/07/20-challenging-job-interview-puzzles-which-every-analyst-solve-at-least **1 / 1 point**

What can you say about this problem?

( ) This problem involves an unavoidable degree of probability or uncertainty in its solution.
( ) This problem requires way too much common-sense knowledge about the behavior of (e.g.) sand inside a bottle.
( ) This problem involves a high degree of mental imagery and thus would be difficult for a computer program to solve.
(•) This problem seems, in fact, amenable to a "standard" problem-space representation, and to solution via search. Occasionally our standard methods work, thank heavens.

✓ **Correct**

7. Consider the following propositional logic statements: **1 / 1 point**

IF (P AND Q) THEN R

P AND S

Q AND V

Which of the following statements cannot be derived from these three statements?

( ) R
( ) P AND R
( ) S AND V
(•) (NOT P) OR (NOT Q)

✓ **Correct**

8. Consider the hammerhead shark. Why do you think the animal evolved this sort of head shape? **1 / 1 point**

https\://www.nationalgeographic.com/content/dam/animals/thumbs/rights-exempt/fish/group/hammerhead-sharks_thumb.ngsversion.1498159813652.adapt.1900.1.jpg

(•) Presumably the distance between the two eyes helps with binocular vision (particularly for points in the region between the two eyes).
( ) It couldn't have had anything to do with binocular vision.
( ) This is a likely instance of "sexual selection", like the peacock's tail: males evolved this head shape because females prefer it.
( ) The head shape is probably a good instance of predatory camouflage.

✓ **Correct**

9. Why do we need to use a toroid shape in the Schelling model of neighborhoods as discussed in the lecture? **1 / 1 point**

(•) A toroid allows all individuals to have exactly 8 neighbors, otherwise the corner location cannot possibly have 4 neighbors.
( ) A toroid is just more interesting - it doesn't actually matter for our neighborhood problem.
( ) A toroid is a better representation of a neighborhood, because neighborhoods aren't usually perfect squares
( ) A toroid is better suited to a multiple-agent model because there are edges.

✓ **Correct**

10. What does this photograph suggest about the difficulty of the "vision" problem? **1 / 1 point**

https://en.wikipedia.org/wiki/Penrose_triangle#/media/File:Perth_Impossible_Triangle.jpg

(•) Vision, because it involves all sorts of guesswork and heuristics in interpreting three dimensions from a two-dimensional projection, is capable of being confused or misled by certain images
( ) Vision is an easier problem for interpreting curved objects than interpreting objects with corners, like the "impossible triangle" in the photo.
( ) We can never get the vision problem "right" so it's not even worth our while to attempt it.
( ) Vision is easy because we can always interpret retinal images unambiguously.

✓ **Correct**